ENCODED INEQUALITY: GENDER BIAS IN TECH JOB LANGUAGE ACROSS SENIORITY LEVELS

Team : Anushka Nayak, Mallerly Mena

GOAL: Investigate whether gender bias in tech job descriptions varies across career seniority levels using computational language analysis.

Hypotheses:

- Job descriptions contain measurable gender-coded language
- Bias strength differs across entry, mid, senior, and leadership roles
- More senior roles may show stronger masculine associations due to leadership stereotypes

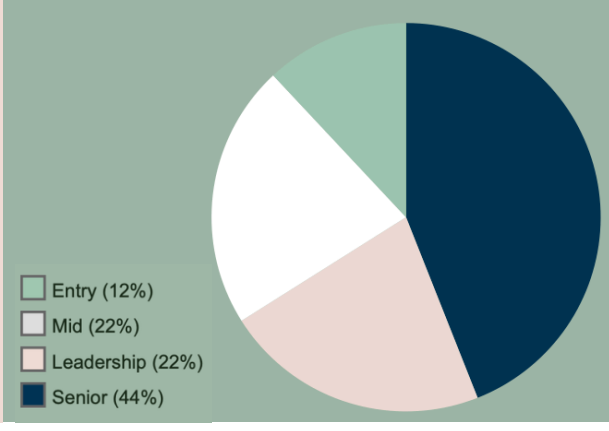
DATASET & CORPUS SCOPE

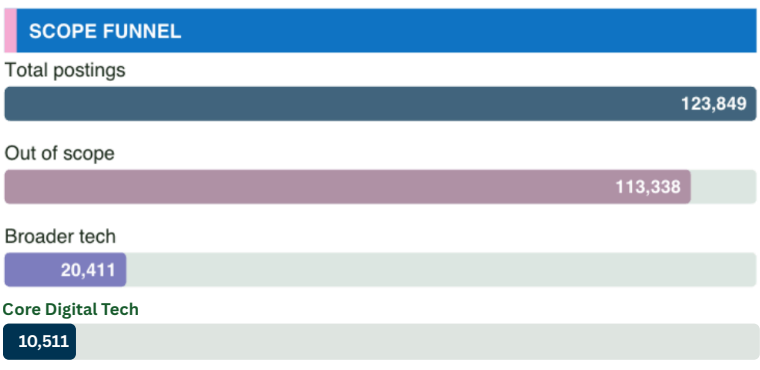
LinkedIn job postings from 2023-2024 were filtered to digital-technology occupations.

- **~124,000** original postings
- ~9,600 documents after cleaning and deduplication
- Seniority groups: entry, mid, senior, leadership
- IT and software industries dominate the corpus

123,849
total postings

LinkedIn US job postings (2024). Scoped to **10,511 digital-tech core** postings.





Data Quality

Many metadata fields were incomplete.

- Salary information missing for most postings
- Skills descriptions frequently absent
- Remote status often unspecified
- Leadership roles underrepresented

These gaps reflect real-world platform data limitations.

EXPERIMENT DESIGN

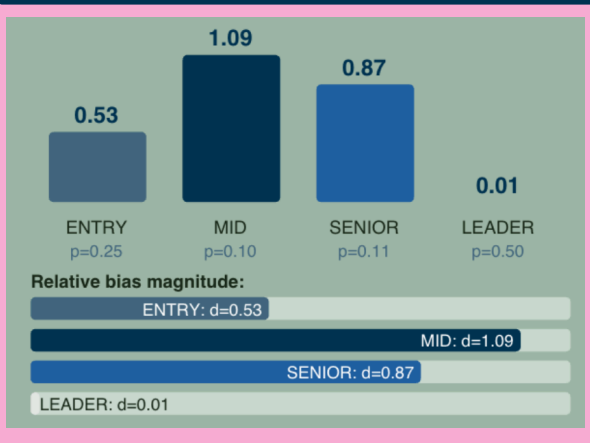
1 CORPUS Digital-tech job descriptions cleaned & deduplicated. 4 seniority groups trained separately.

2 EMBEDDINGS FastText word2vec model per seniority group, 3 random seeds (7, 17, 29) for stability.

3 WEAT TEST Word Embedding Association Test: cosine similarity of gender target words vs. career-attribute sets.

4 MONOTONICITY Check if bias effect sizes change monotonically: entry → mid → senior → leadership.

WEAT EFFECT SIZES BY SENIORITY



Monotonicity Not Supported

Bias does not increase consistently with seniority. Mid-level roles exhibit the strongest gender associations rather than leadership positions.

Mid-level: strongest bias
$d \approx 1.09$, $p \approx 0.10$ — highest observed effect size across groups.

Entry & Senior: moderate bias
$d \approx 0.5–0.9$ — detectable associations, but not statistically significant.

Leadership: minimal bias
$d \approx 0.01$, $p \approx 0.50$ — little to no measurable gender association.

Statistical significance
None of the results reach conventional significance ($p < 0.05$).

Interpretation
Findings indicate directional trends only. Limited sample sizes reduce statistical power, particularly for leadership roles.

ROBUSTNESS CHECKS

Labeled-only vs. Backfilled

Effect size directions consistent across both label strategies (backfill does not introduce bias).

Deduped vs. Non-deduped

Removing ~1,800 duplicate postings yields similar WEAT effect sizes — dedup did not distort results.

Broader Tech slice (20,411)

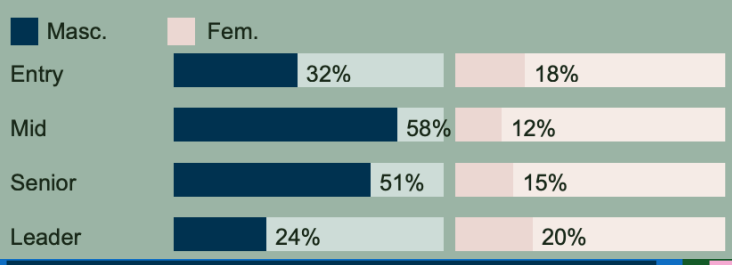
Larger company-scoped corpus mirrors digital-tech core patterns across all four seniority groups.

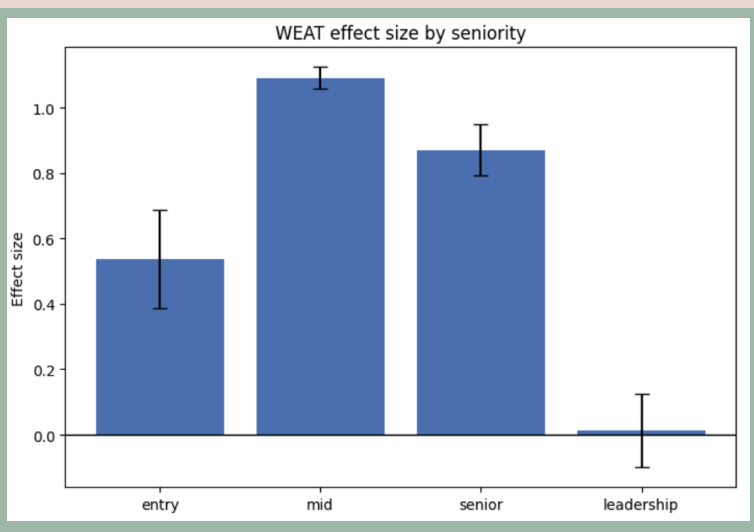
Lexicon-based gender audit

Masculine-coded terms cluster in mid & senior descriptions; feminine terms sparse throughout all levels.

GENDER-CODED LANGUAGE

Masculine-coded terms (e.g. *competitive, dominant, analytical*) cluster in **mid & senior** descriptions. Feminine-coded terms (e.g. *collaborative, supportive*) appear sparingly, slightly elevated at entry level.





CONCLUSION

↑ **Bias IS present in tech job postings**

≠ **Pattern is NOT monotonic across levels**

▲ **Mid-level postings are most gendered**

✓ **Robustness checks confirm consistency**

LIMITATIONS & FUTURE WORK

- Small leadership corpus ($n \approx 58$ docs) limits statistical power.
- No WEAT tests reach $p < 0.05$ — results are suggestive, not conclusive.
- FastText on small corpora may be unstable despite multi-seed averaging.
- Future work: scale leadership data, test alternative attribute word sets, compare pre-trained embeddings.